

Nguyễn Thiện Nhân - 2458879 - HW2

TEKO
SOLUTION

Date:

1. $H = -p_1 \log_2 p_1 - p_2 \log_2 p_2$

A: (12, 0) →

→ $p_1 = \frac{12}{12} = 1, p_2 = 0$

→ $H = -1 \cdot \log_2(1) = 0$

B: (9, 3)

$p_1 = \frac{9}{12} = 0.75, p_2 = \frac{3}{12} = 0.25$

→ $H = -0.75 \log_2(0.75) - 0.25 \log_2(0.25) = 0.811$

C: (6, 6)

$p_1 = p_2 = 0.5$

→ $H = -0.5 \log_2(0.5) - 0.5 \log_2(0.5) = 1$

⇒ $A < B < C$

2. Gini impurity = $1 - \sum_k p_k^2$

→ Weighted Gini = $\frac{n_L}{n} \cdot \text{Gini}_L + \frac{n_R}{n} \cdot \text{Gini}_R$

S1: (6P, 1N), (5P, 6N)

Child 1: $G_1 = 1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2 = 0.245$

Child 2: $G_2 = 1 - \left(\frac{5}{11}\right)^2 - \left(\frac{6}{11}\right)^2 \approx 0,496$

$\rightarrow$ $Gini(S_1) = \frac{7}{18} \cdot 0,245 + \frac{11}{18} \cdot 0,496$

$\approx 0,398$

S2: (8P, 4N) (3P, 3N)

Child 1: $G_1 = 1 - \left(\frac{8}{12}\right)^2 - \left(\frac{4}{12}\right)^2 \approx 0,444$

Child 2: $G_2 = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 0,5$

$\rightarrow Gini(S_2) = \frac{12}{18} \cdot 0,444 + \frac{6}{18} \cdot 0,5$

$= 0,463$

$\rightarrow$ CART picks S1 (0,398 < 0,463)

3. 4 7 9 13 18 21 29
 P P P N N N N

Try cut points: 5,5 ; 11 ; 19,5

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5,5: Split P | P P N N N N

$$G_L = 0$$

$$G_R = 1 \cdot \left(\frac{2}{6}\right)^2 + \left(\frac{4}{6}\right)^2 = \frac{4}{9}$$

$$\rightarrow \text{Gini}(5,5) = \frac{1}{7} (0) + \frac{6}{7} \left(\frac{4}{9}\right) \approx 0,381$$

11: Split: P P P | N N N N

$$G_L = 0$$

$$G_R = 0$$

$$\rightarrow \text{Gini}(11) = 0$$

19,5: Split: P P P N N | N N

$$G_L = 1 \cdot \frac{3^2}{5} + \frac{2^2}{5} = 0,48, \quad G_R = 0$$

$$\rightarrow \text{Gini}(19,5) = \frac{5}{7} \cdot 0,48 + \frac{2}{7} \cdot 0$$

$$\approx 0,343$$

$\rightarrow$ Pick 11.

4. Original subtree: 7/30 validation error

Replace subtree by 1 leaf: 6/30 error

$\rightarrow$ we should prune the subtree and replace with leaf.

Reduced-error pruning keep the change if it doesn't make validation performance worse. It's even lower in this case $\rightarrow$ more simple, better generalization.

2 strategies for handling missing data:

1. Impute missing feature value like replacing with median.
 - $\rightarrow$ May lose information if the missing data is meaningful.
2. Send missing observations to a default branch with more training samples.
 - $\rightarrow$ Simpler, no need to guess missing value.
 - $\rightarrow$ May wrong branch because the most popular one might not fit.

5. HW2:

I will change max-depth for smaller. I expect training score to decrease because shallower tree has less capacity to fit training data. However, validation score may improve if original tree is overfitting.

	baseline	depth = 5
depth	20	5
leaves	136	16
train macro-f1	1.000	0.9257
val macro-f1	0.8373	0.8227
top 3	+ Gravity Acc - min() - X,	
feature	+ Body Acc, Jerk - max() - X,	
	+ Gravity Acc - max() - X.	

Train macro-f1 fell as expected but val macro-f1 also fell.
→ depth 5 too restrictive and increase bias.